

Figures and Tables

The Results section: part 1

Fish 290

Tuesday Feb 4

Reading

We will spend the next 3 class periods on the Results section, including figures and tables.

Our time in class is limited.

The readings on Canvas* are chosen to fill in gaps we might not cover in class.

*readings from *The Scientist's Guide to Writing* by Stephen Heard

Presentation of Results:

Today's agenda:

1. **When to use a figure, table, or neither?**
2. What options do I have for figures?
3. How to make an *effective* figure or table?

Presentation of Results:

Three complementary approaches

- 1) **Text:** We state in words what we found - what happened.
 - We will cover text in more detail later
- 2) **Figures:** Graphs help **visualize** complex relationships and draw the reader's attention to the most important findings.
- 3) **Tables:** Provide **details** on the results that are important

Most journals encourage “supplemental” or “appendix” tables for more details. Journals also encourage electronic storage of raw data for public access (this is a new feature of science).

For your studies, including raw data as a table is Fine.

Presentation of Results:

Three complementary approaches

Not every result needs a figure or table. Too many graphics will dilute your main points and confuse your reader. If everything is important than nothing is.

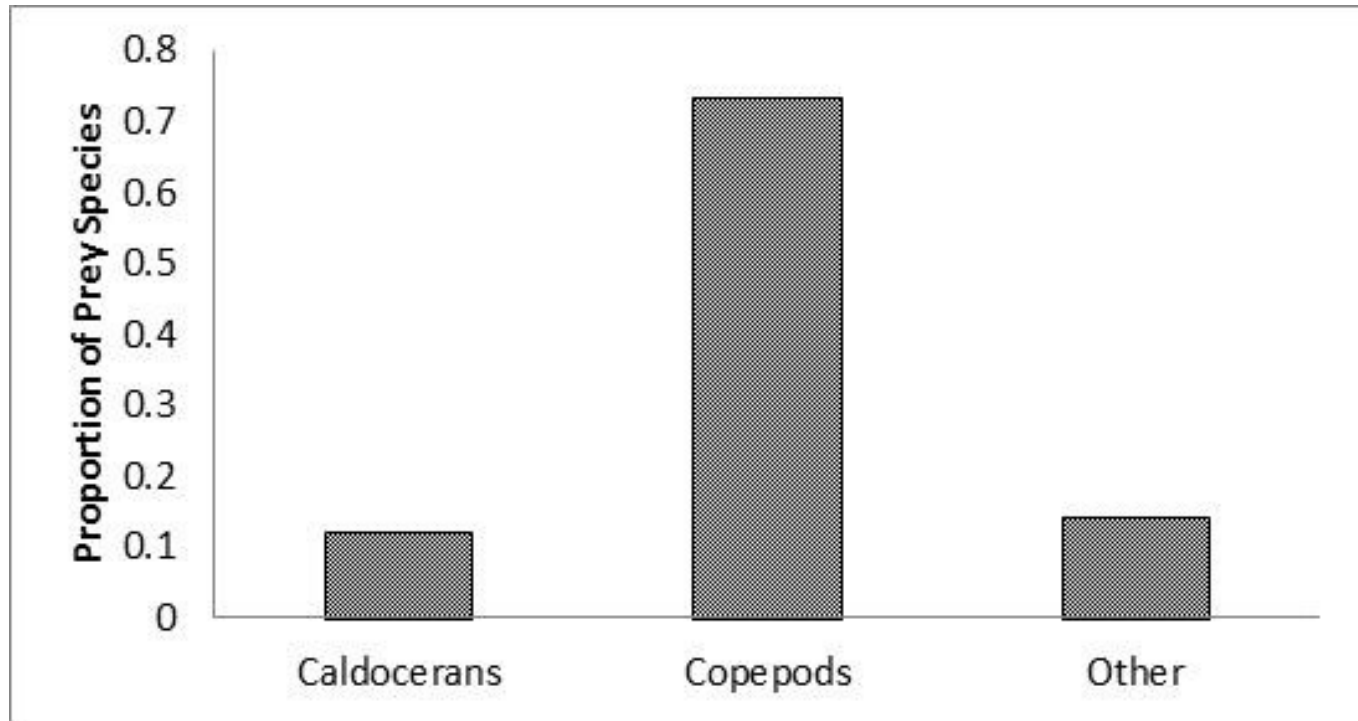
When do I use a figure?

When do I use a table?

When do I not bother with either and just write a sentence?

When to use just a sentence:

Remember that graphics compete for your reader's attention. Use a sentence or two for simple results with few numbers.



How many data points are shown in this figure?

When to use just a sentence:

Remember that graphics compete for your reader's attention. Use a sentence or two for simple results with few numbers.

“Copepods comprised the highest proportion of prey items (75%). Caldocerans comprised 12% of prey items, and the remaining 13% of items were unidentified.”

Do you need a figure to understand this result?

When to use a graphic (either figure or table):

Try this sentence on for size:

“For lakebottom sediments collected at the inflow, average grain size was 1.3 mm (± 0.4 mm standard deviation). Smaller grains predominated farther out: average 0.8 mm (± 0.2 mm) at 5 m from the inflow, 0.2 mm (± 0.004 mm) at 40 m, and 0.002 mm (± 0.0005 mm) at 60 m.”

UGH SO COMPLICATED!!!!

When to use a graphic (either figure or table):

Try this sentence on for size:

“For lakebottom sediments collected at the inflow, average grain size was 1.3 mm (± 0.4 mm standard deviation). Smaller grains predominated farther out: average 0.8 mm (± 0.2 mm) at 5 m from the inflow, 0.2 mm (± 0.004 mm) at 40 m, and 0.002 mm (± 0.0005 mm) at 60 m.”

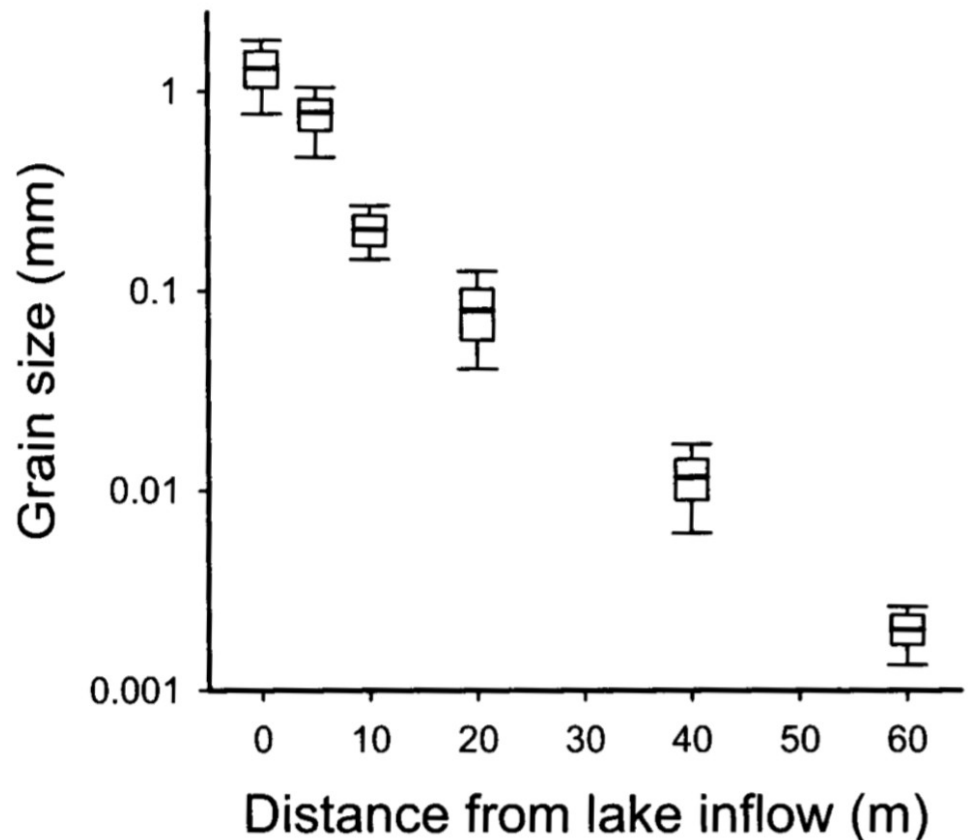
Numbers are inherently demanding to read.

Use a graphic to display complicated relationships or numerical information.

When to use a graphic (either figure or table):

“Grain size was large at the inflow and decreased farther out (Fig. 1).”

A breath of fresh air!



When to use a Figure versus a Table

Figures are best at showing *relationships* and *trends*.

Think: what do I want the reader to take away from this graphic?

- If it's a **trend** (an increase or decrease) or a relationship (two variables interact) then use a **figure**.

When to use a Figure versus a Table

Figures are best at showing *relationships* and *trends*.

However, tables are good at showing *precise numbers* and *multiple variables at once*.

Think: what do I want the reader to take away from this graphic?

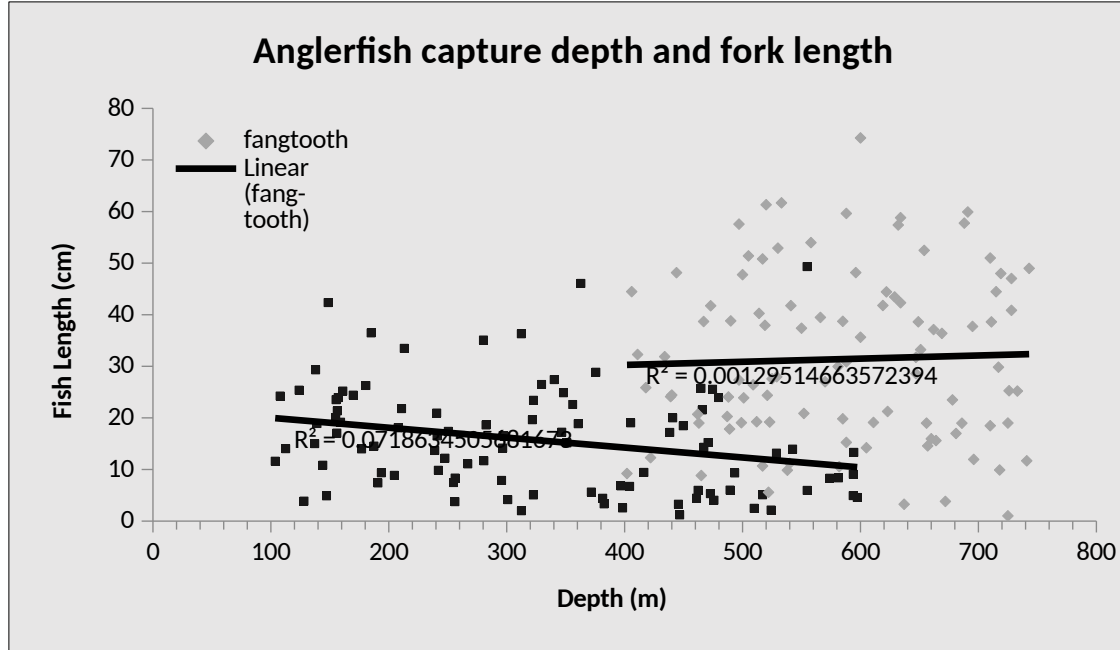
- If it's a **trend** (an increase or decrease) or a relationship (two variables interact) then use a **figure**.
- If it's the **exact value** (like a p-value) and there are too many entries for one sentence, then use a **table**.

Presentation of Results:

Today's agenda:

1. When to use a figure, table, or neither?
2. **What options do I have for figures?**
3. How to make an *effective* figure or table?

Types of figures: scatterplots



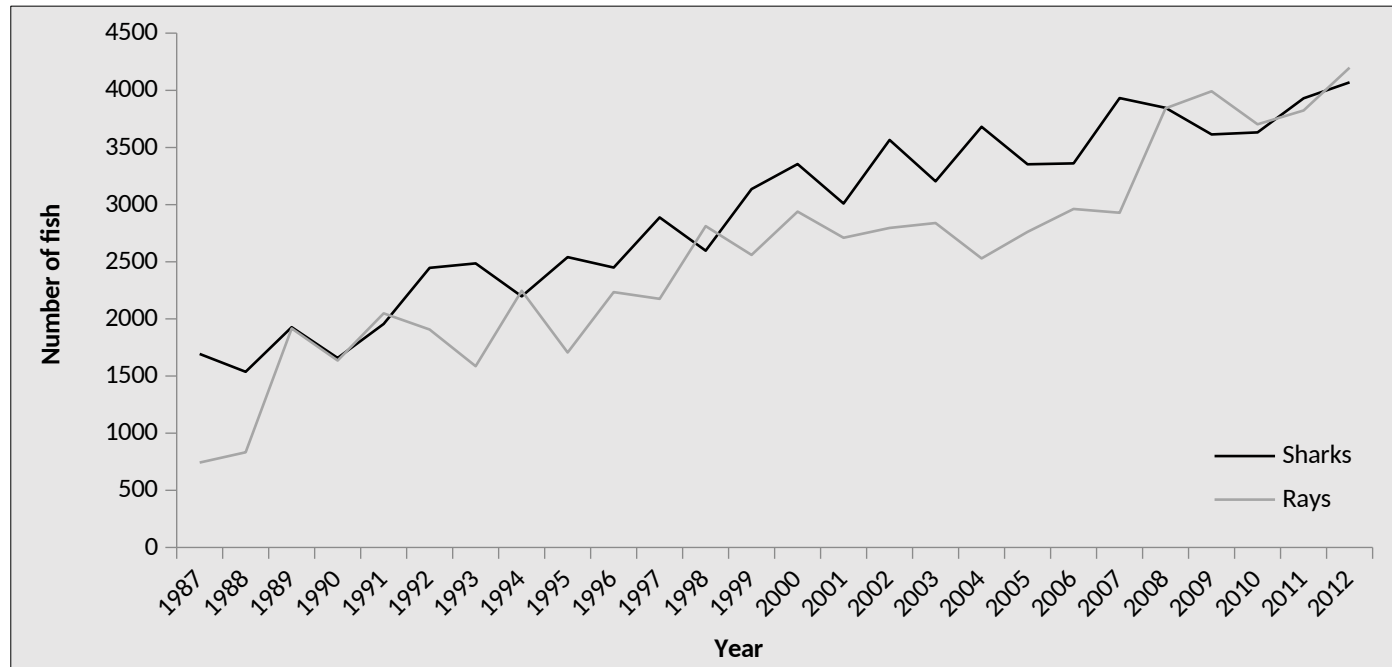
Attributes:

- Shows relationship between data
- Independent data (x axis) influences dependent data (y axis)
- Good when both variables are continuous data
- Can show trends
- Shows exact data points, not summary statistics (what is a summary statistic?)

Warnings:

- Both variables MUST be continuous
- Make sure multiple series on graph are comparable
- Do not have too many different data sets

Types of figures: line plots



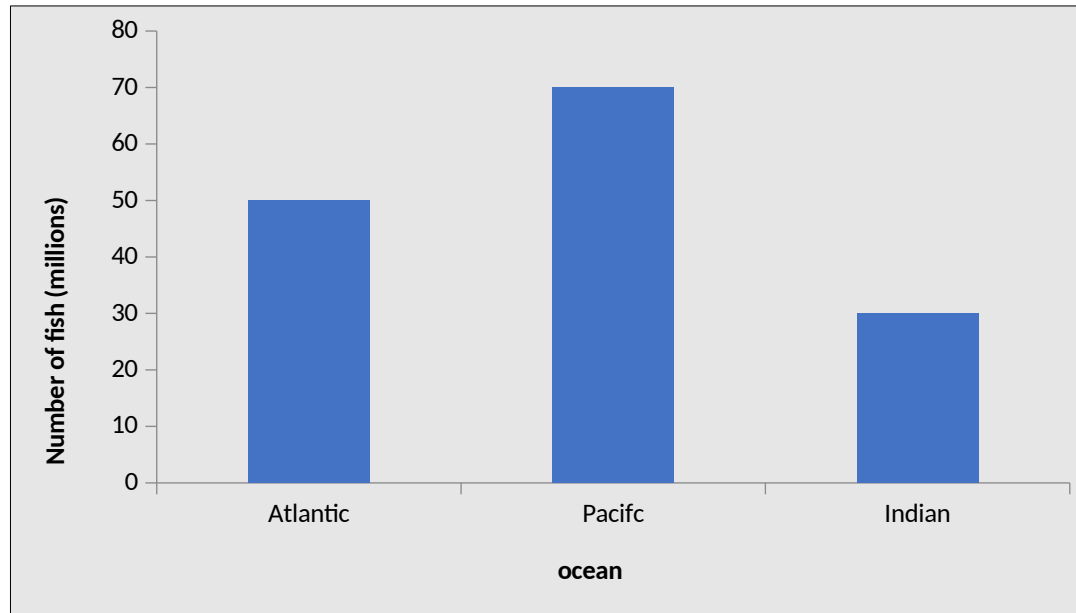
Attributes:

- Good if one variable is continuous and one is numerical but discrete (e.g. time)
- Can compare multiple series of data
- Suggests values between data points
- Patterns are readily apparent

Warnings:

- Data must be continuous!
- For each line, one datum per observation
- Intermediate values must be valid
- Poor choice if few data points
- Make sure multiple series on graph are comparable (same scale)

Types of figures: bar graphs



Numbers of fish caught per year in commercial fisheries in the Atlantic, Pacific, and Indian oceans.

Attributes:

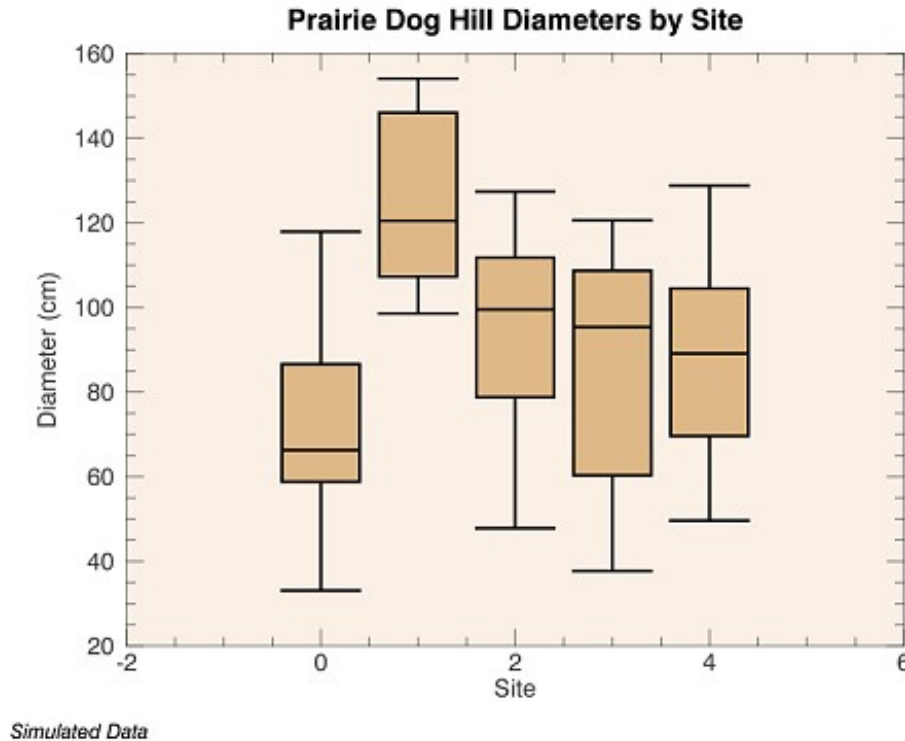
- Each bar represents a discrete data point
- Patterns are visually striking
- The order of the bars can reveal patterns
- Can compare categories

Warnings:

- Not great for continuous data
- May not reveal variation around the mean or bar value
- Bars order can suggest false trends
- Takes up a *lot* of space to show a few data points- use wisely

Types of figures: boxplots (box and whisker)

Boxplots: shows data spread including median, quartiles and outliers



Attributes:

- Bars show variation, including summary statistics
- Patterns are visually striking
- The order of the bars can reveal patterns
- Can compare categories

Warnings:

- Bars order can suggest false trends
- “whiskers” can mean different things (standard deviation, vs quartiles vs. total range, etc.)

Other types of figures to consider

- **Maps, photographs, or flowcharts:** Usually does not show data or results. Instead, used to give the reader a sense of what you did (Methods)
- **Histogram:** uses bars to show counts of one variable.
- **Pie chart*:** shows proportions of a whole

* The case for/against pie charts: <https://www.geckoboard.com/blog/pie-charts/>

Presentation of Results:

Today's agenda:

1. When to use a figure, table, or neither?
2. What options do I have for figures?
3. **How to make an *effective* figure or table?**

Making effective figures

There are many possible ways to display the same result.

Just because a program can make it,
doesn't mean it's correct or the best option.

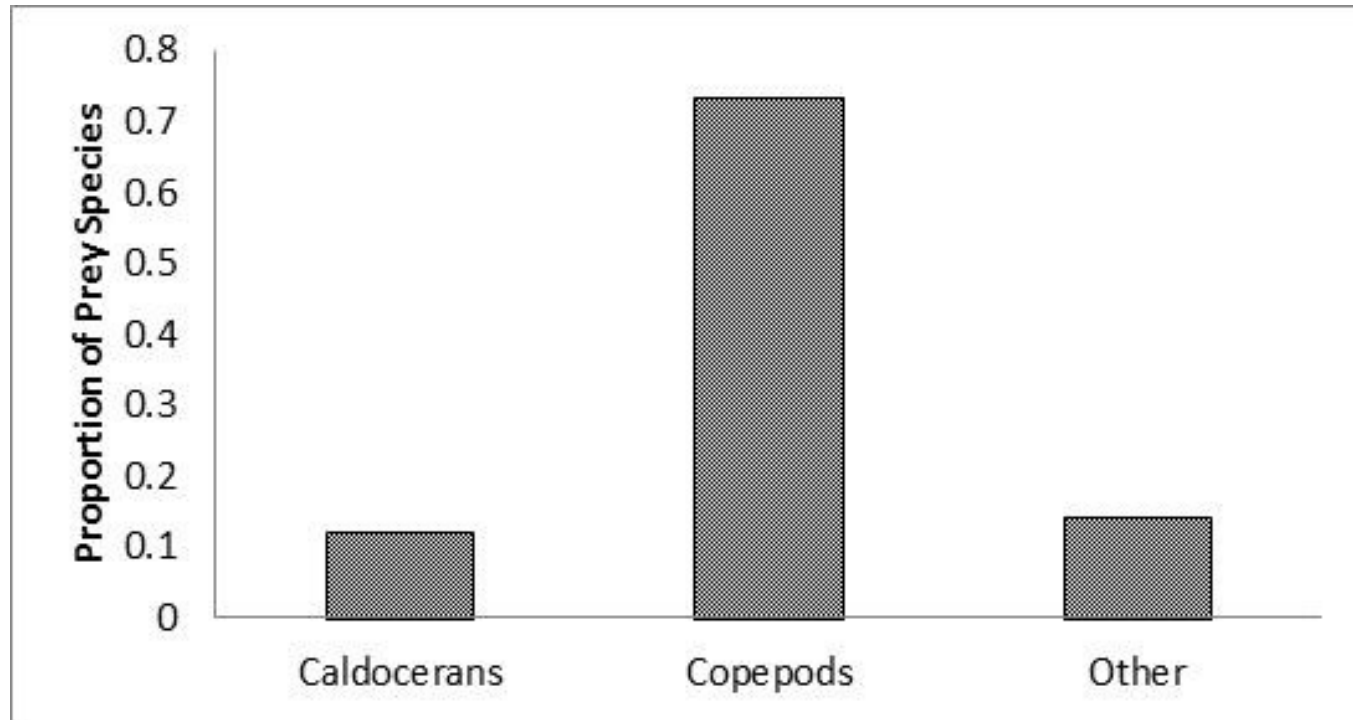
The golden rule of figures and table design:

Give readers the **most information capable of being digested** in the **shortest time** with the **least ink** in the **smallest space** possible.

Making effective figures

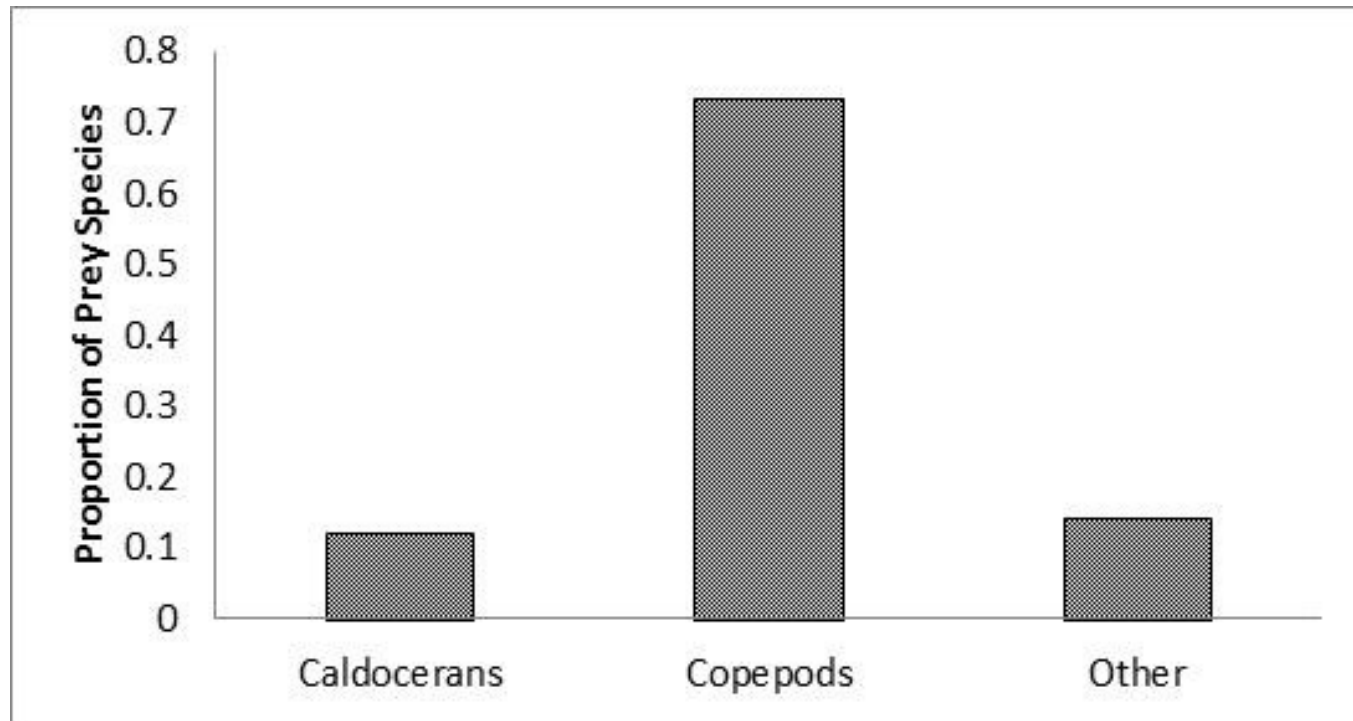
Critique the following figures. Avoid critiquing captions for now (we'll come back to that)

Making effective figures



Graph showing average prey proportions in Lake Washington. Calculated from data collected from 1993-2011.

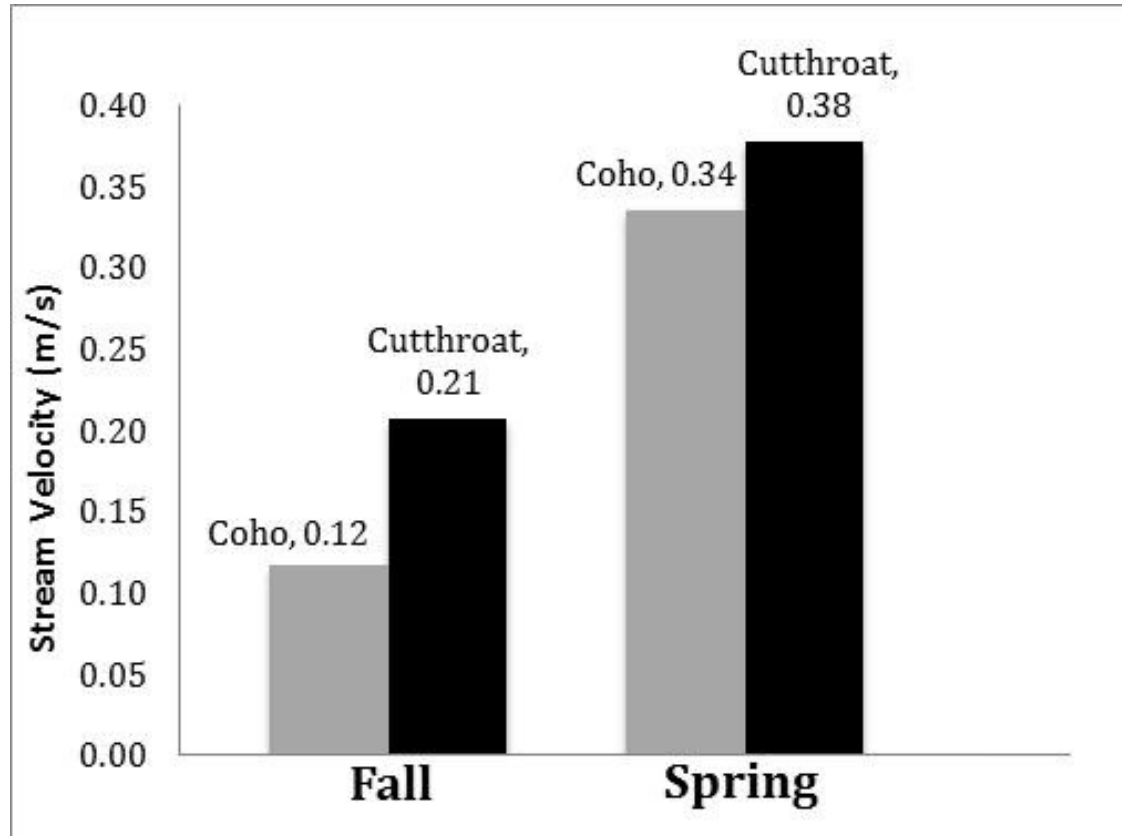
Making effective figures



Graph showing average prey proportions in Lake Washington.
Calculated from data collected from 1993-2011.

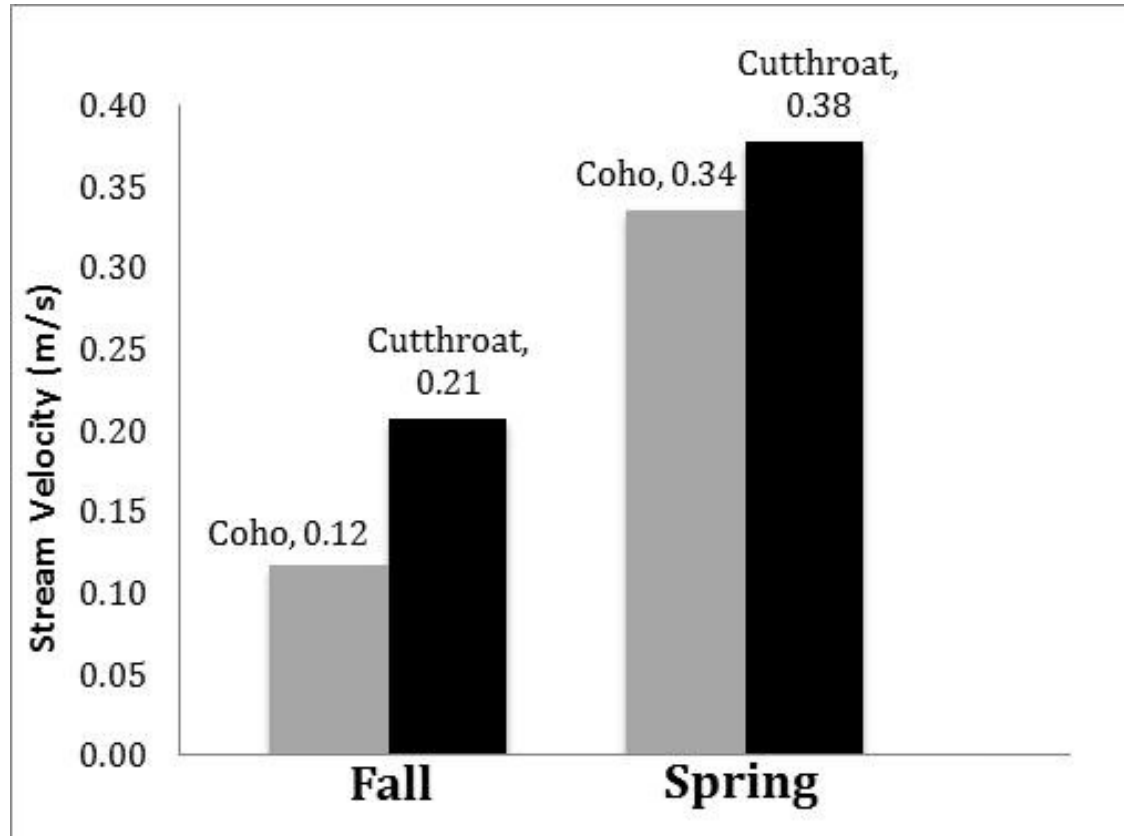
Critique: This figure contains only three data points: the proportions of the three prey species. This can easily be expressed in a single sentence.

Making effective figures



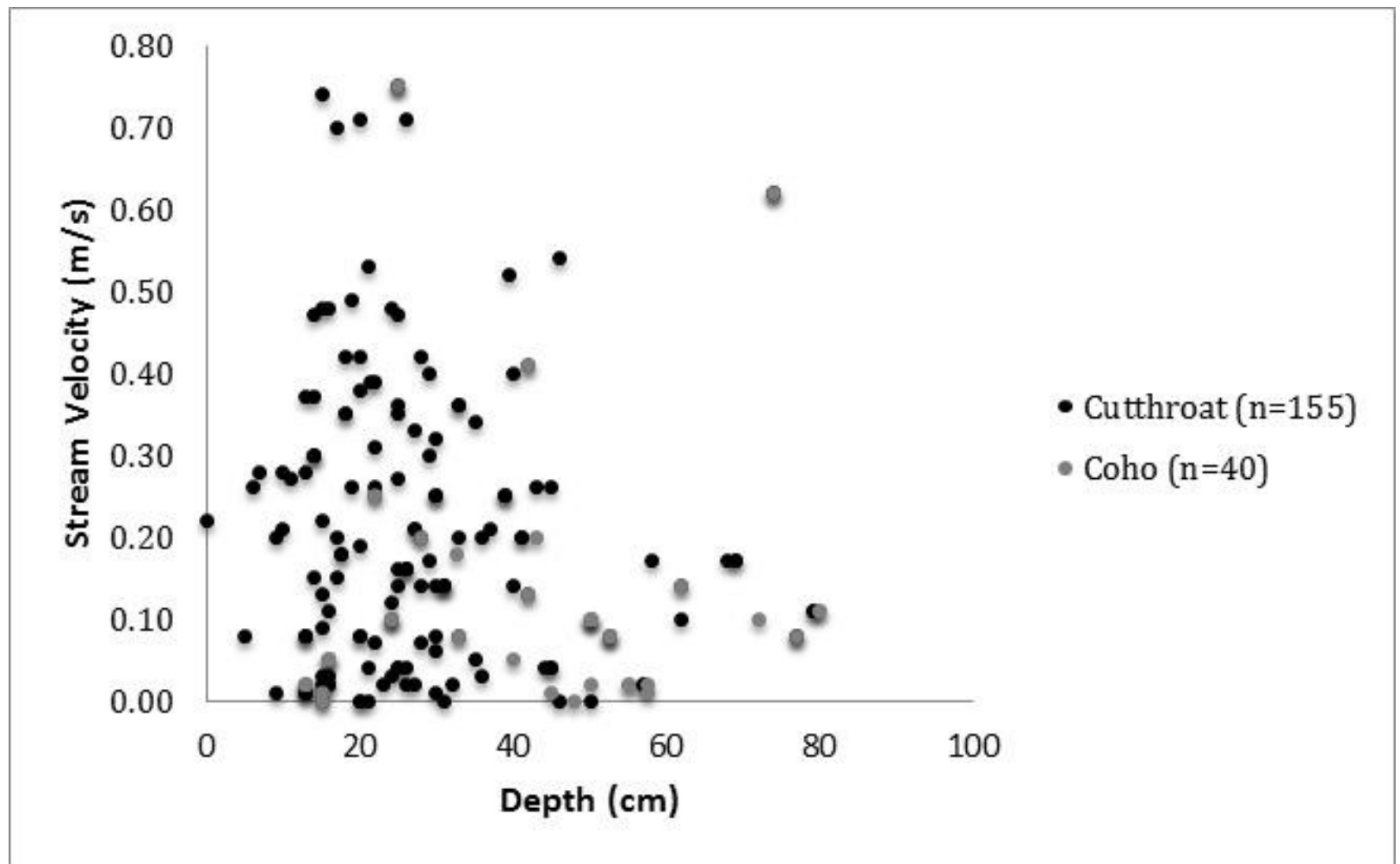
Average stream velocity of habitat where fish were captured in Rock Creek

Making effective figures

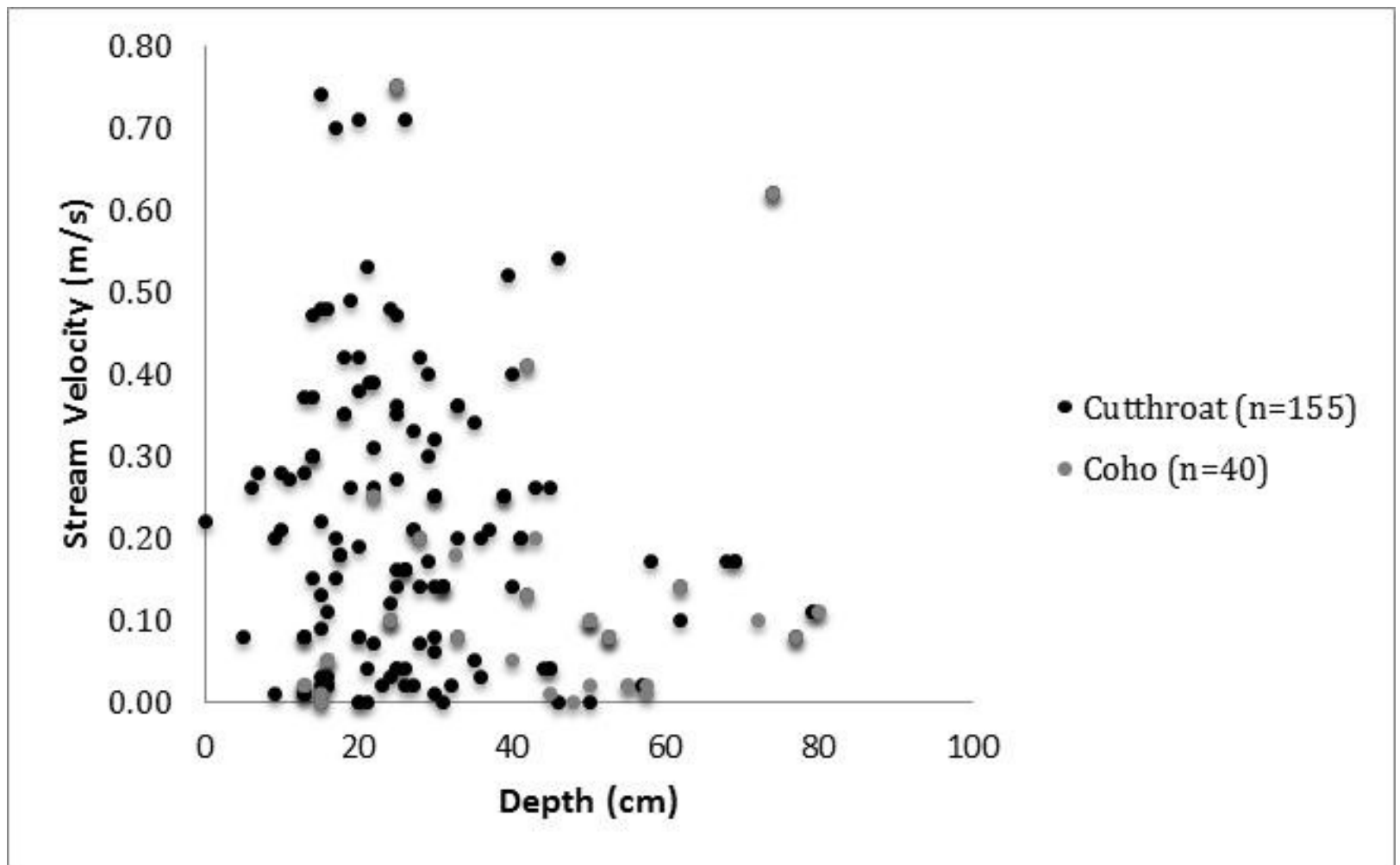


Average stream velocity of habitat where fish were captured in Rock Creek

Critique: Better, but the figure has the data in numerical form and this is redundant with the height of the bars. In addition, there is no indication of variation around the means.



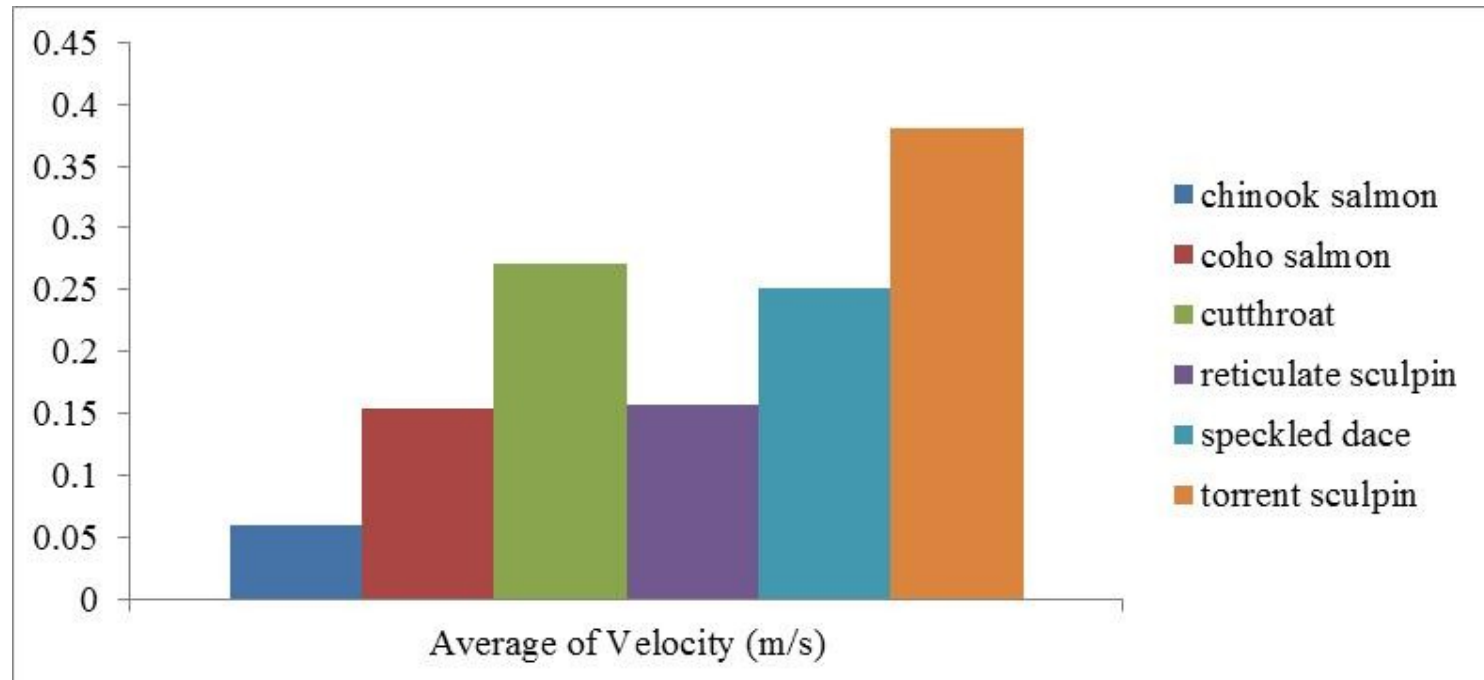
Stream velocity vs. depth of habitat occupied by captured fish in Rock Creek (fall)



Stream velocity vs. depth of habitat occupied by captured fish in Rock Creek (fall)

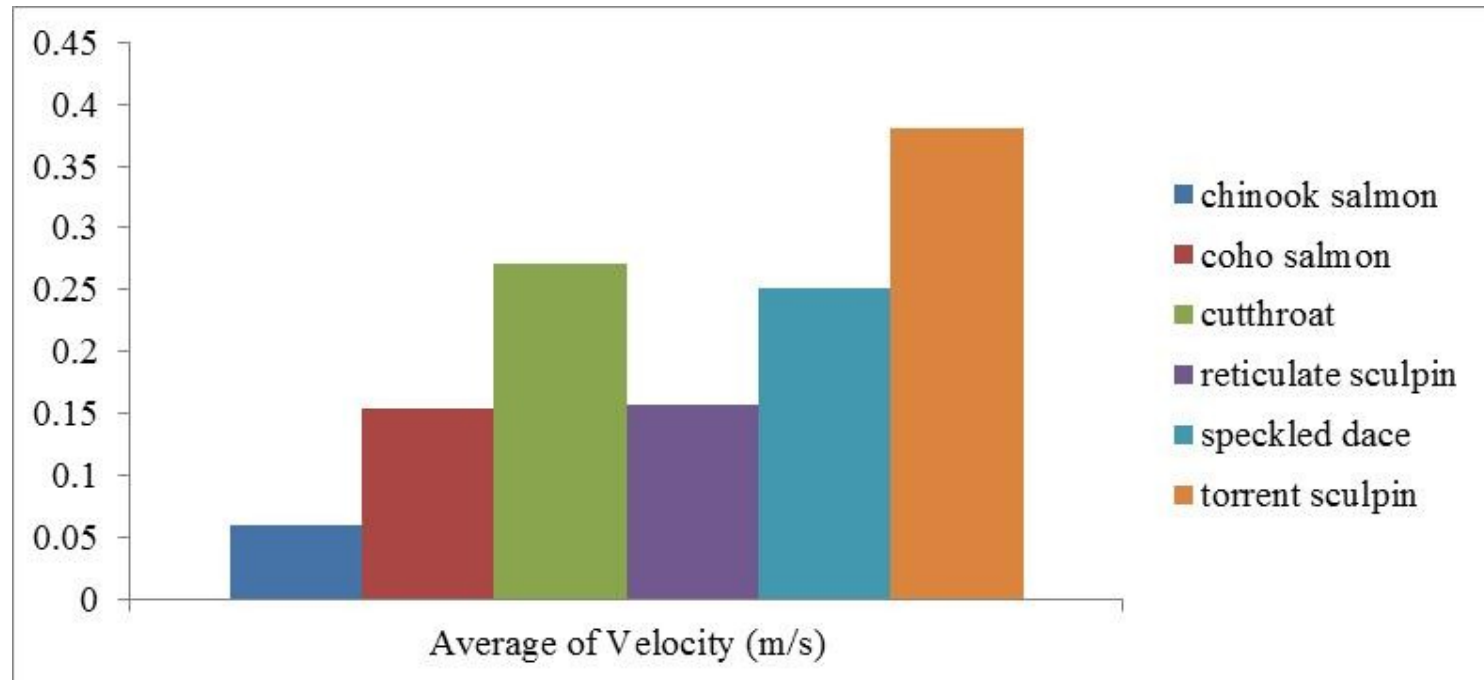
Critique: This is better, but the two sets of points need more distinct symbols to tell them apart.

Making effective figures



The average velocity (m/s) of where the instream species were found.

Making effective figures



The average velocity (m/s) of where the instream species were found.

Critique: The x-axis has the label that belongs on the y-axis. Why are the species listed in this order? It is alphabetical but so what? What is the message here? Also, no variation among means shown.

Making effective figures

The figures you just saw violate parts of the golden rule:

Give readers the **most information***
capable of being digested in the **shortest**
time with the **least ink** in the **smallest space**
possible.

*incorrect information doesn't count.

Do not mislead!

Assignments for today and for home

In class:

1. Work on practice data to get used to making graphs in Excel
2. Draw out ideas for figures and tables for your own data
3. Make figures and tables for your own data

At home:

4. Read “The Results Section” on Canvas
5. Continue data analysis/graphics on your own
6. Complete **peer reviews** by end-of-day **Thursday**
7. Continue working on your Intro+Methods (**Methods due Friday**)

Next time in class, we will cover Tables and Captions.